

## **Healthcare Cost and Expenditure Analysis**

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## **Introduction**

Healthcare expenditure analysis plays an important role in understanding medical cost patterns, identifying high-risk patient groups, and supporting data-driven healthcare decision-making. Factors such as age, smoking habits, BMI, and geographic region can significantly influence healthcare spending. This project focuses on analyzing healthcare insurance cost data to identify demographic cost drivers and expenditure trends using Python-based analytics and visualization techniques.

## **Objective**

The objective of this project was to analyze healthcare expenditure data to identify high-cost patient patterns, demographic spending variations, and regional healthcare cost trends through exploratory data analysis and visualization.

## **Dataset Information**

The dataset used for this project was the Medical Cost Personal Dataset obtained from Kaggle. The dataset contained healthcare insurance and medical expenditure information for 1,338 individuals.

The dataset included the following variables:

- Age
- Gender

- BMI
- Number of Children
- Smoking Status
- Region
- Medical Charges

## **Tools and Technologies**

The following tools and technologies were used in this project:

- Python
- Pandas
- Matplotlib
- Jupyter Notebook

## **Methodology**

### **Data Loading and Exploration**

The dataset was loaded into Python using Pandas. Exploratory data analysis was conducted to understand healthcare expenditure distributions, demographic patterns, and healthcare cost drivers.

### **Cost Trend Analysis**

Healthcare charges were analyzed across:

- Smoking status

- Geographic region
- Age groups
- BMI trends

## **Data Visualization**

Multiple visualizations were developed using Matplotlib to identify healthcare expenditure patterns and support business-oriented healthcare insights.

## **Results and Findings**

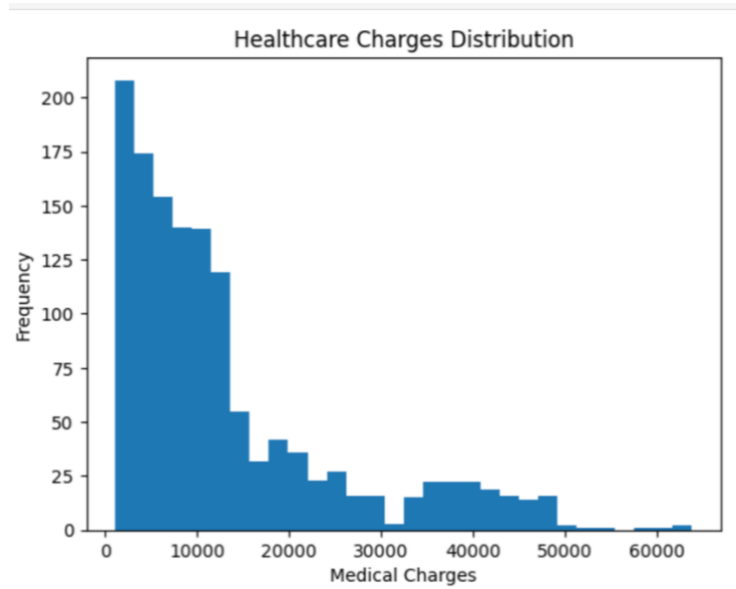
Key findings from the analysis included:

- Smokers had significantly higher medical charges compared to non-smokers
- The southeast region showed the highest average healthcare expenditure
- Healthcare costs generally increased with patient age
- The average BMI exceeded 30, indicating obesity-related healthcare risk trends
- Healthcare expenditure distribution was highly right-skewed due to high-cost patient populations

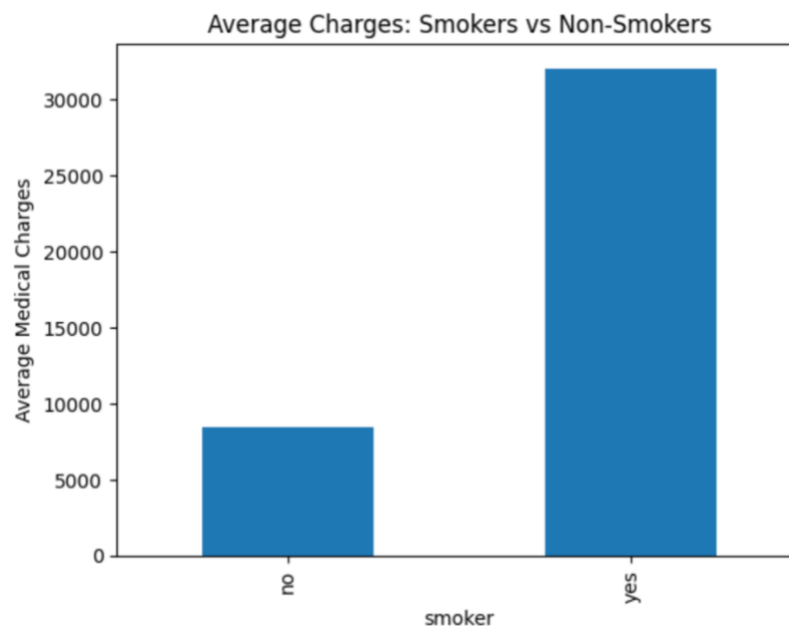
The analysis demonstrated how demographic and lifestyle factors can significantly impact healthcare spending patterns.

## **Visualizations**

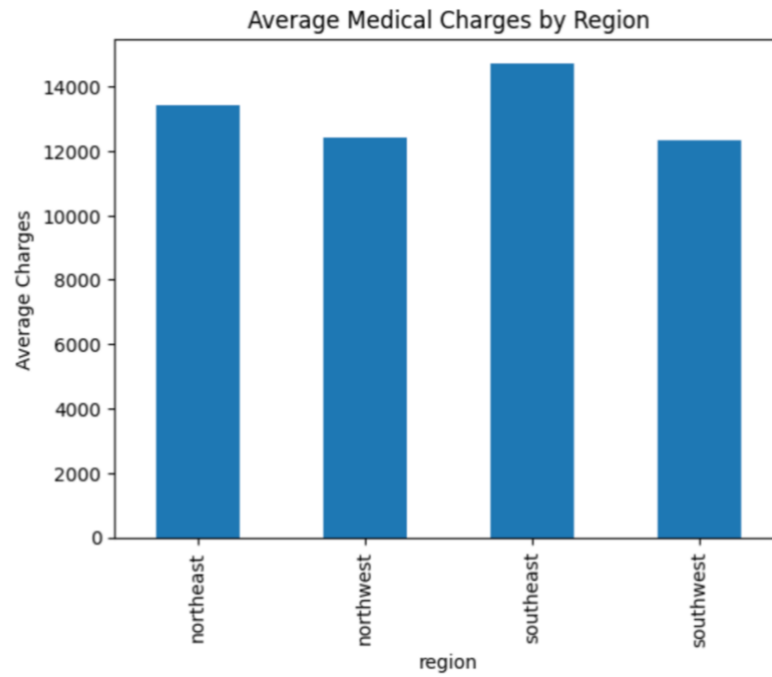
## Healthcare Charges Distribution



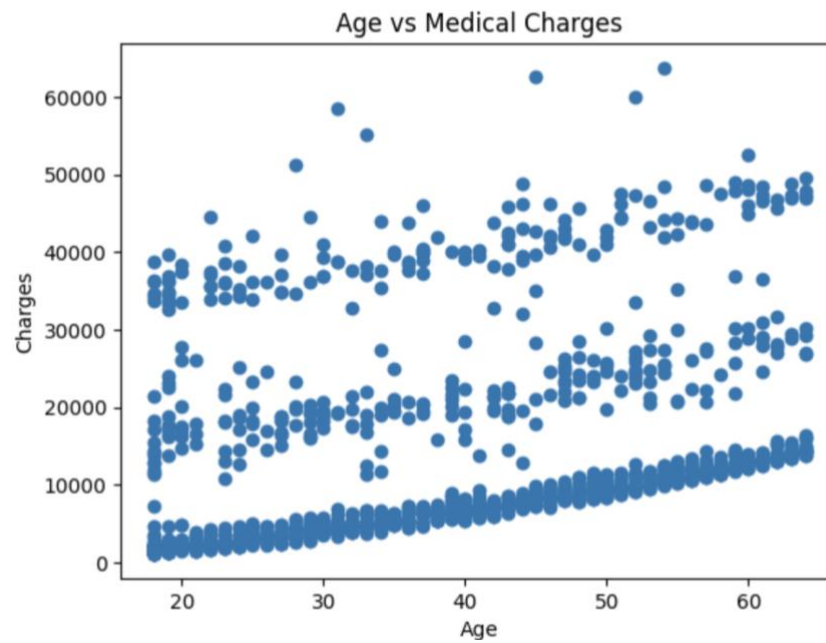
## Smoker vs Non-Smoker Cost Comparison



## Regional Healthcare Charges



## Age vs Medical Charges



## **Conclusion**

This project demonstrated how healthcare expenditure data can be analyzed using Python to identify demographic spending trends and high-cost healthcare patterns. Through exploratory analysis and data visualization, the project provided practical experience in healthcare business analytics, expenditure analysis, and data-driven reporting. The findings highlight the importance of lifestyle and demographic factors in influencing healthcare costs and support the use of analytics for healthcare decision-making.